ShotStream: Streaming Multi-Shot Video Generation for Interactive Storytelling

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Research Questions

- How can multi-shot video generation be reformulated to enable interactive, streaming storytelling with real-time user input?
- How can causal autoregressive models maintain visual consistency across multiple shots while avoiding error accumulation?
- What distillation strategy can effectively transfer bidirectional multi-shot generation capabilities to fast causal models?

Core Idea

Current multi-shot video generation models use bidirectional attention to process entire sequences at once, making them slow and non-interactive. ShotStream reformulates this as a causal next-shot prediction problem, where each shot is generated conditioning on previous shots. The key innovation is a dual-cache memory system: a global cache stores sparse frames from historical shots for inter-shot consistency, while a local cache holds recent frames within the current shot for intra-shot consistency. A RoPE discontinuity indicator helps distinguish between these caches. The model is trained via a two-stage distillation from a bidirectional teacher, progressively moving from ground-truth to self-generated conditioning.

Key Formula

RoPE Discontinuity Indicator

$$\Theta_t = \phi t + k\theta$$

For the t-th latent within k-th shot, Θ_t is temporal rotation angle, φ is base temporal frequency, θ is phase shift representing shot-boundary discontinuity. This explicitly decouples global and local contexts by introducing discrete temporal jumps at shot boundaries.

Context Frame Encoding

$$z_{context} = \varepsilon(V_{context})$$

V_context are sampled sparse context frames from historical shots, ε is the 3D VAE encoder, z_context are the resulting conditioning latents with dimensions f_context×c×h×w. This encodes visual context for next-shot generation.

Key Figure

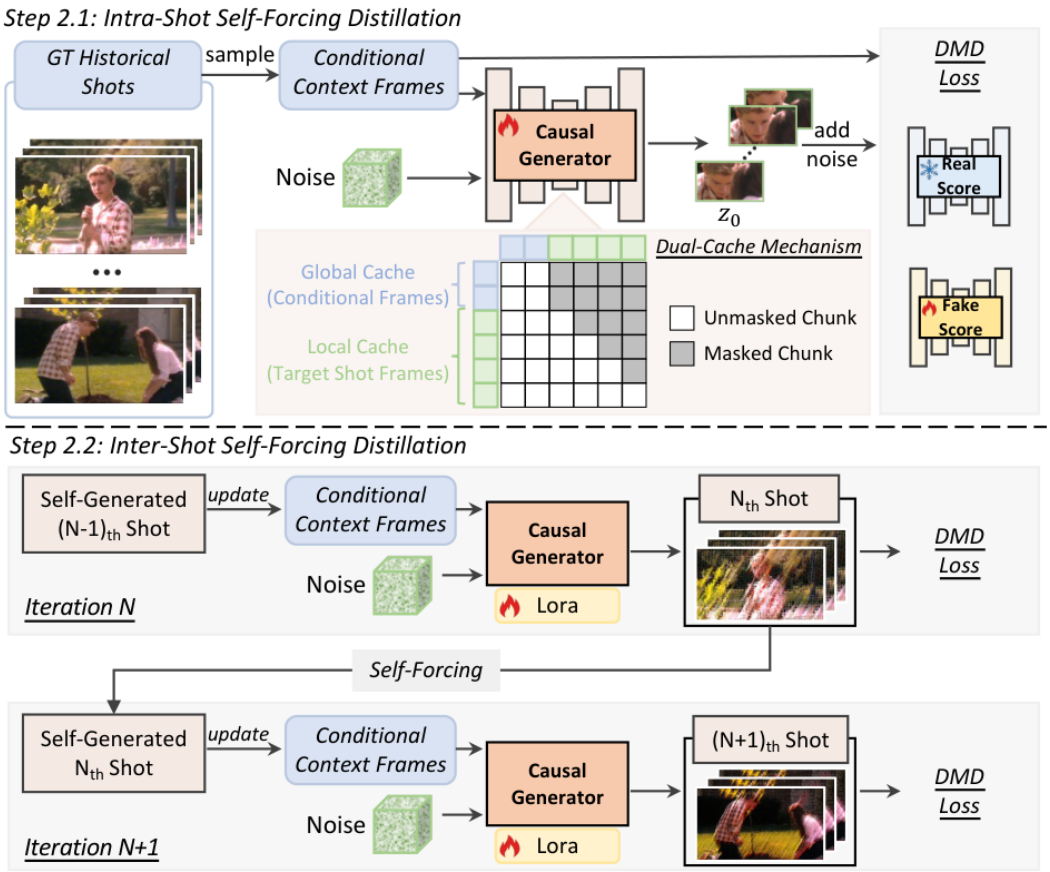

Figure 4. Causal Architecture and Two-Stage Distillation Pipeline. We distill a slow, multi-step bidirectional teacher into an efficient, few-step causal generator. To maintain visual coherence, we propose a novel dual-cache memory mechanism: a global context cache stores conditional frames to ensure inter-shot consistency, while a local context cache retains generated frames within the target shot to guarantee intra-shot consistency. To prevent error accumulation, we employ a progressive two-stage distillation strategy. In the first stage, intra-shot self-forcing distillation (Step 2.1), the model is conditioned on ground-truth historical shots to causally generate the current shot chunk-by-chunk. In the second stage, inter-shot self-forcing distillation (Step 2.2), the model is conditioned on its own previously generated shots, rolling out the video shot-by-shot while iteratively generating the frames of each individual shot chunk-by-chunk.

Figure 4: Shows the complete causal architecture and two-stage distillation pipeline, illustrating the dual-cache mechanism and progressive training strategy that are core to the method.

Limitations

- Visual artifacts and inconsistencies when scenes and text prompts are highly complex due to limited backbone model capacity
- Still has room for acceleration to provide better interactive experiences despite 16 FPS performance
- Dynamic sampling strategy for historical frames may not capture optimal context for all narrative scenarios
- Two-stage distillation adds training complexity compared to single-stage approaches